

EX 3 – Demonstrate missing value analysis, fixing missing values and outlier analysis in dataset

AIM :

To demonstrate missing value analysis, methods for handling missing data, and outlier detection in a dataset using R/Python.

ALGORITHM :

1. Missing Value Analysis

1. Import required libraries (Pandas, NumPy)
2. Load the dataset
3. Identify missing values using functions like `isnull()` or `is.na()`
4. Count total missing values in each column
5. Visualize missing data (optional using heatmaps/plots)

2. Fixing Missing Values

1. If missing values are few → remove rows/columns
2. If important data → replace missing values using:
 - Mean (for numerical data)
 - Median (for skewed data)
 - Mode (for categorical data)
3. Use forward fill or backward fill if needed
4. Verify that missing values are handled

3. Outlier Analysis

1. Detect outliers using:
 - Boxplot method
 - Z-score method
 - IQR (Interquartile Range) method
2. Identify data points far from normal range
3. Handle outliers by:

- Removing extreme values
 - Replacing with median/mean
 - Capping (limiting values to a threshold)
4. Visualize results using boxplots or scatter plots

CODE :

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

data={
    'Age':[22,25,np.nan,30,28,np.nan,45,120],
    'Salary':[25000,30000,28000,np.nan,32000,31000,100000,29000],
    'Experience':[1,3,2,5,np.nan,40,4,3]
}

df=pd.DataFrame(data)

print("original dataset:\n",df)
print("Missing Value Count:\n",df.isnull().sum())

df['Age'].fillna(df['Age'].mean(),inplace=True)
df['Salary'].fillna(df['Salary'].median(),inplace=True)
df['Experience'].fillna(df['Experience'].mode()[0],inplace=True)

print("\nDataset after handling missing values :\n",df)

Q1=df.quantile(0.25)
```

```

Q3=df.quantile(0.75)

IQR=Q3-Q1

outliers=((df<(Q1-1.5*IQR))|(df>(Q3+1.5*IQR)))

print("\nOutliers detected:\n",outliers)

df_cleaned=df[~outliers.any(axis=1)]

print("\nDataset after handling outliers:\n",df_cleaned)

df_cleaned.boxplot()

plt.title("Boxplot after outliers removal")

plt.show()

```

OUTPUT :

```

original dataset:
   Age  Salary  Experience
0  22.0  25000.0         1.0
1  25.0  30000.0         3.0
2   NaN  28000.0         2.0
3  30.0     NaN         5.0
4  28.0  32000.0         NaN
5   NaN  31000.0        40.0
6  45.0 100000.0         4.0
7 120.0  29000.0         3.0
Missing Value Count:
Age      2
Salary   1
Experience 1
dtype: int64

Dataset after handling missing values :
   Age  Salary  Experience
0  22.0  25000.0         1.0
1  25.0  30000.0         3.0
2  45.0  28000.0         2.0
3  30.0  30000.0         5.0
4  28.0  32000.0         3.0
5  45.0  31000.0        40.0
6  45.0 100000.0         4.0
7 120.0  29000.0         3.0

Outliers detected:
   Age  Salary  Experience
0  False  False     False
1  False  False     False
2  False  False     False
3  False  False     False
4  False  False     False
5  False  False      True
6  False   True     False
7   True  False     False

Dataset after handling outliers:
   Age  Salary  Experience
0  22.0  25000.0         1.0
1  25.0  30000.0         3.0
2  45.0  28000.0         2.0
3  30.0  30000.0         5.0
4  28.0  32000.0         3.0

```

RESULT :

Thus the above program is executed and the result is obtained.